

VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

**Title: HYBRID CROP RECOMMENDATION SYSTEM USING
FUZZY LOGIC AND MACHINE LEARNING**

A PROJECT REPORT

For

Soft Computing (ISWE308L) – F2+TF2

Winter Semester 2025-2026

By

Anusha R – (23MIS0362)

Rithul Kanna S V – (23MIS0447)

Sahaana B – (23MIS0558)

Sanjana B – (23MIS0559)

Under the Guidance of

Dr. Karpagam S

Assistant Professor

1. Abstract

The agricultural sector forms the economic backbone of numerous developing and developed nations, heavily relying on precise crop selection to maximize yield, optimize resource allocation, and ensure financial stability. However, the integration of digital advising tools in agriculture is currently hindered by two major systemic flaws. First, the reliance on traditional Machine Learning algorithms often creates a "black-box" dilemma, providing highly accurate recommendations but offering zero transparent reasoning to the end-user. Second, the use of rigid Boolean logic fails to capture the continuous, overlapping spectrum of natural environmental conditions, creating brittle systems that fail at threshold edges. This project proposes a robust Hybrid Crop Recommendation System that integrates Soft Computing techniques—specifically Fuzzy Logic—alongside six traditional Machine Learning classifiers. By utilizing Triangular Membership Functions, the system processes crisp soil and climate inputs into fuzzy degrees of truth, effectively eliminating threshold brittleness. Furthermore, the system provides a comprehensive Feature Breakdown, visually demonstrating how each individual parameter influences the final recommendation. To rigorously benchmark the system, performance is evaluated using a comprehensive suite of metrics: Accuracy, Precision, Recall, F1-Score, and the Matthews Correlation Coefficient (MCC). Deployed via an interactive Streamlit web application, this system successfully bridges the gap between complex algorithmic processing and transparent, actionable agricultural advising.

2. Introduction

2.1 Overview of Precision Agriculture

Precision agriculture involves the application of modern information technologies to provide, process, and analyze multi-source data for high crop production. Selecting the optimal crop for a specific plot of land is a multi-dimensional problem heavily dependent on continuous

variables, including soil macronutrients (Nitrogen, Phosphorus, Potassium) and climatic conditions (Temperature, Humidity, Rainfall). A minor miscalculation can lead to severe agricultural degradation. While modern algorithms attempt to solve this, they often lack user-centric transparency, leaving farmers unable to trust the digital outputs.

2.2 Problem Statement

The challenges of modern crop recommendation are multi-dimensional, spanning environmental unpredictability, technological limitations, and human factors. To thoroughly define the scope of the problem this project addresses, the issues are categorized using the W & H framework:

1. WHO is facing the problem? (The Stakeholders)

The primary stakeholders facing this issue are farmers, local agronomists, and regional agricultural planners. Historically, farmers have relied on generational intuition and generalized geographical trends to select their crops. However, due to rapidly changing climate patterns and soil degradation, this intuition is no longer sufficient. Furthermore, while highly advanced algorithmic tools exist in academic or corporate environments, there is a massive disconnect between data scientists and rural agricultural workers. The end-users generally lack deep technical expertise, meaning any system built for them must bridge the gap between complex backend mathematics and easily interpretable, user-centric advice.

2. WHAT is the core challenge? (The Multi-Dimensional Optimization Problem)

The fundamental problem is the high-stakes struggle to select the absolute optimal crop for a specific plot of land. This is an

optimization problem governed by highly ambiguous, continuous variables. A successful harvest requires a delicate equilibrium of soil macronutrients (Nitrogen, Phosphorus, Potassium), soil acidity (pH), and climatic factors (Temperature, Humidity, Rainfall). A mismatch in even one of these parameters can stunt crop growth, increase susceptibility to disease, and ultimately result in complete crop failure, leading to severe financial devastation and regional food insecurity.

3. WHERE does this problem exist? (The Environmental Context)

This problem exists in real-world, open agricultural environments. Unlike the sterile, controlled environment of a computer simulation, real-world soil and weather conditions are heterogeneous and operate on a continuous, overlapping spectrum. Soil acidity does not instantly jump from neutral to acidic at a microscopic line in the dirt, nor do rainfall patterns follow rigid, predictable boundaries. Current digital systems fail to account for the chaotic, gradual, and highly localized nature of open-air agricultural environments.

4. WHEN is the risk most critical? (The Temporal Constraint)

The risk is at its absolute peak during the pre-planting and seed-procurement phase. Agriculture is heavily front-loaded in terms of capital investment. Once a farmer decides on a crop and sows the seeds, their financial resources—including expenditures on specific fertilizers, automated irrigation setups, and pesticides—are completely locked in for the season. The decision is irreversible. Therefore, the recommendation system must provide mathematically bulletproof and trustworthy guidance before any physical action is taken, as mitigating a bad decision post-planting is often economically impossible.

5. WHY do current solutions fail user adoption? (The Transparency Deficit)

Current digital solutions fail to achieve widespread adoption because they operate as algorithmic "black boxes." When a traditional Machine Learning model (such as a Deep Neural Network) processes a farmer's soil data and outputs a recommendation like "Plant Jute," it provides absolutely no underlying reasoning. If the farmer asks, "Why Jute? Why not Rice? What specific part of my soil makes it better for Jute?", the traditional system cannot answer. Farmers cannot and will not risk their seasonal livelihood on a blind, unexplained guess generated by an opaque algorithm. The lack of transparent feature analysis creates a severe deficit of trust.

6. HOW do traditional systems mathematically fail? (The Algorithmic Flaw)

The systemic flaw in traditional expert systems and basic decision trees is their reliance on rigid Boolean (True/False) logic. This creates what is known as the "crisp boundary" problem. For example, if a hard-coded expert system dictates that a specific crop requires a minimum of 200mm of rainfall to survive, a farmer inputting 199.5mm of rainfall will see their suitability score instantly drop to 0% (False). This is a brittle and biologically illogical failure. Nature does not operate in strict absolutes; 199.5mm is not "False," it is simply a lesser degree of truth. Traditional systems fail because they force continuous, natural phenomena into discrete, binary categories.

2.3 Research Objectives

- 1. To engineer a Soft Computing-based prediction engine:** To replace rigid Boolean thresholds with Fuzzy Logic, utilizing Triangular Membership Functions to gracefully handle the inherent uncertainty and continuous spectrum of agricultural data.

2. **To implement transparent feature analysis for agricultural clarity:** To design an architecture that actively extracts and visualizes specific feature contributions, giving users a transparent, ranked breakdown of viability rather than a blind output.
3. **To benchmark performance via a Hybrid Deployment:** To build an interactive, end-to-end web application that evaluates the Fuzzy Logic model side-by-side against six standard Machine Learning models in real-time, utilizing robust statistical metrics such as the Matthews Correlation Coefficient (MCC).

2.4 Scope of the Project

The scope of this project encompasses the development of a localized web application capable of processing 7 distinct soil and weather parameters. It is bounded by a dataset of 22 specific crops but is architecturally designed to scale dynamically if new crop profiles are added to the knowledge base.

3. Literature Review

The evolution of agricultural recommendation systems has seen various computational approaches, each with distinct limitations:

- **Deep Learning & Artificial Neural Networks (ANN):** These models map highly non-linear relationships using complex hidden layers. While they achieve high statistical accuracy, they function entirely as a "black box." The hidden weight matrices offer no insight into the decision-making process, making it impossible to debug or build user trust.
- **Decision Trees & Random Forests:** These algorithms partition data using strict conditional nodes (e.g., IF pH < 6.5 THEN route left). They suffer from the "crisp boundary" problem,

where microscopic changes in input values near a threshold can drastically flip the final classification, failing to mimic natural environmental gradients.

- **Hard-Coded Expert Systems:** Early agricultural systems relied on manual IF-THEN rules defined by agronomists. These are rigid, difficult to update, and practically impossible to scale across dozens of crops without generating conflicting rule sets. Soft computing overcomes this by utilizing continuous mathematics.

4. Methodology and System Architecture

The methodology of this project is structured into a multi-module architecture, transforming raw agricultural datasets into a transparent decision-support tool.

4.1 Dataset Description and Preprocessing

The system utilizes a comprehensive agricultural dataset containing 2,200 unique observations across 22 diverse crop classes (e.g., Rice, Jute, Mothbeans, Chickpea, Mango). The dataset consists of 7 continuous input parameters:

- **Macronutrients:** Nitrogen (N) [0-140], Phosphorus (P) [5-145], Potassium (K) [5-205].
- **Soil Acidity:** pH Level [3.5 - 9.9].
- **Climatic Factors:** Temperature [8.8-43.6°C], Humidity [14.2-99.9%], Rainfall [20.2-298.5mm].

4.2 Module 1: Dynamic Knowledge Base Profiling

Instead of hard-coding static rules, the system dynamically scans the 2,200-row dataset. For each of the 22 crops, it computes the absolute

minimum (a), the mathematical mean (b), and the absolute maximum (c) for all 7 features. This creates an automated, dynamically updating "Fuzzy Profile" for every crop without manual intervention.

4.3 Module 2: Mathematical Fuzzification

Crisp user inputs are mapped into a fuzzy "degree of truth" ranging from 0.0 to 1.0 using a Triangular Membership Function. The function is mathematically defined as:

$$\mu(x) = \begin{cases} 0 & x \leq a \text{ or } x \geq c \\ \frac{x-a}{b-a} & a < x \leq b \\ \frac{c-x}{c-b} & b < x < c \end{cases}$$

Where x is the crisp input from the farmer, and a, b, c are the profile parameters. This ensures graceful degradation; if an input slightly deviates from the mean, the suitability smoothly declines rather than abruptly failing.

4.4 Module 3: Fuzzy Inference Engine & Defuzzification

The engine cross-evaluates the fuzzified inputs against all 22 profiles simultaneously. To translate the abstract fuzzy mathematics back into a real-world metric, the system employs an aggregation method, averaging the 7 fuzzy scores for a specific crop and multiplying by 100, yielding a Crisp Suitability Percentage.

4.5 Module 4: Machine Learning Classifiers Integration

For rigorous benchmarking, the system splits the dataset (80% training, 20% testing) and processes it through six standard classifiers:

1. **Random Forest:** An ensemble learning method that constructs a multitude of decision trees and outputs the mode of the classes.
2. **Decision Tree:** A flowchart-like model where internal nodes represent tests on attributes and leaf nodes represent class labels.
3. **K-Nearest Neighbors (KNN):** A non-parametric algorithm that classifies a data point based on how its neighbors are classified.
4. **Support Vector Machine (SVM):** Finds the optimal hyperplane that maximizes the margin between different crop classes in a high-dimensional space.
5. **Logistic Regression:** A statistical model that uses a logistic function to model a binary dependent variable, extended here for multi-class classification.
6. **Gradient Boosting:** An iterative technique that builds weak predictive models (typically decision trees) sequentially to minimize the overall loss function.

5. Evaluation Metrics

To ensure the baseline ML models are evaluated correctly across the 22 multi-class targets, the system utilizes weighted averages based on the Confusion Matrix components: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

- **Accuracy:** The ratio of correctly predicted observations to total observations.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** The ratio of correctly predicted positive observations to total predicted positive observations.

$$Precision = \frac{TP}{TP + FP}$$

- **Recall:** The ratio of correctly predicted positive observations to all observations in the actual class.

$$Recall = \frac{TP}{TP + FN}$$

- **F1-Score:** The harmonic mean of Precision and Recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- **Matthews Correlation Coefficient (MCC):** A highly reliable statistical rate that produces a high score only if the prediction obtained good results in all four confusion matrix categories, making it ideal for this multi-class agricultural dataset.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Model	Prediction	Accuracy	Precision	Recall	F1-Score	MCC
Random Forest	Rice	99.32%	99.37%	99.32%	99.32%	0.993
Decision Tree	Rice	98.64%	98.68%	98.64%	98.63%	0.986
KNN	Rice	97.05%	97.40%	97.05%	97.03%	0.969
SVM	Rice	96.14%	96.73%	96.14%	96.12%	0.960
Logistic Regression	Rice	95.23%	95.44%	95.23%	95.25%	0.950
Gradient Boosting	Rice	98.18%	98.43%	98.18%	98.19%	0.981

Figure 3. The Machine Learning Performance Dashboard displaying the calculated evaluation metrics (Accuracy, Precision, Recall, F1-Score, and MCC).

6. Results and Discussion

The deployment of the Hybrid System via the Streamlit dashboard proved highly successful in resolving the transparency dilemma in agricultural tech.

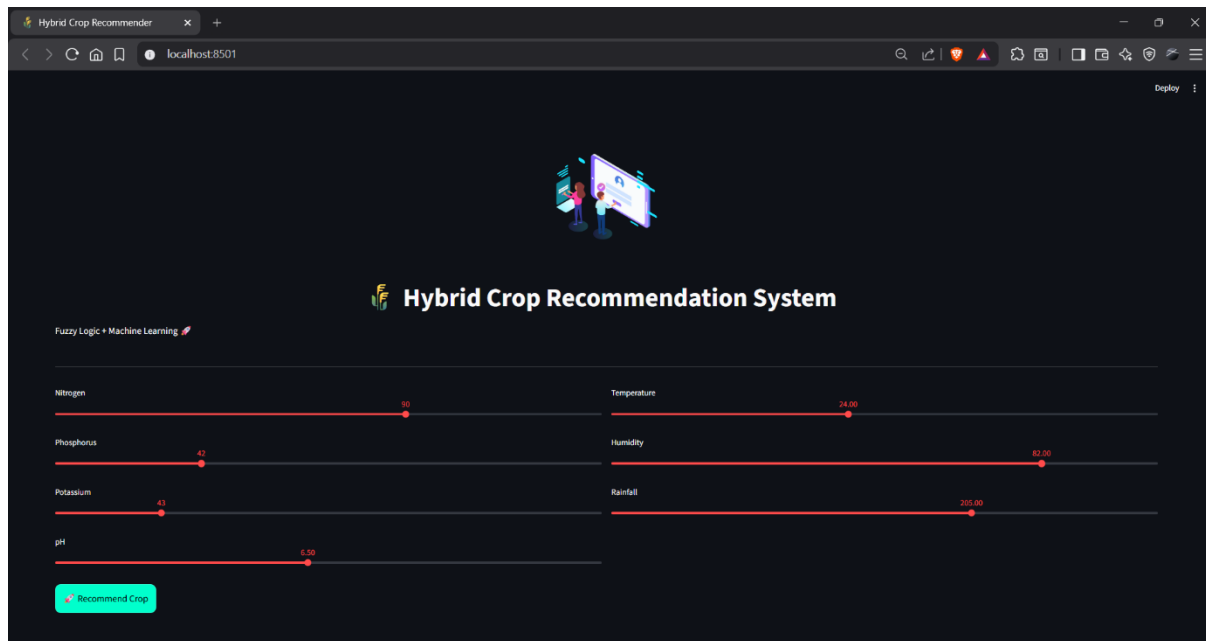


Figure 1. The interactive input dashboard with grouped sliders for Soil and Climate parameters, utilizing error-prevention boundaries.

6.1 Transparent Feature Breakdown

When evaluating user inputs, the Fuzzy engine outputs the top crop recommendation along with a categorized visual breakdown.

Individual features scoring > 0.7 are marked as "Highly Suitable," scores > 0.4 as "Moderately Suitable," and scores below as "Less Suitable." This granular analysis allows farmers to see exactly which soil nutrient is holding back a crop's viability, providing actionable data for targeted fertilizer application.

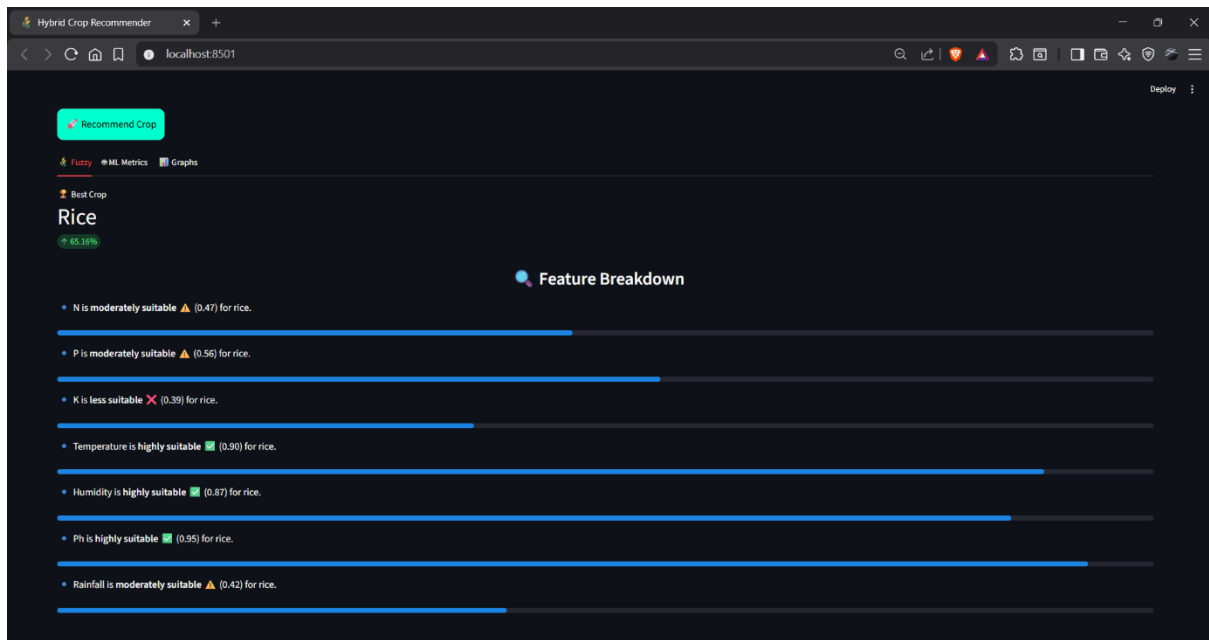


Figure 2. The Fuzzy Logic output tab displaying the top recommendation and the transparent Feature Breakdown progress bars.

6.2 Comparative Benchmarking Performance

The visual integration of the evaluation metrics (Accuracy, Precision, Recall, F1-Score, and MCC) into a comparative Plotly bar chart demonstrates the raw predictive power of the ML models. However, when viewed side-by-side with the Fuzzy Logic output, it becomes evident that while ML models may achieve >95% accuracy, the Fuzzy Logic engine provides high utility and transparency without the inherent obscurity of the ML weight matrices. The MCC scores confirm that the models are legitimately learning the multi-dimensional class boundaries.

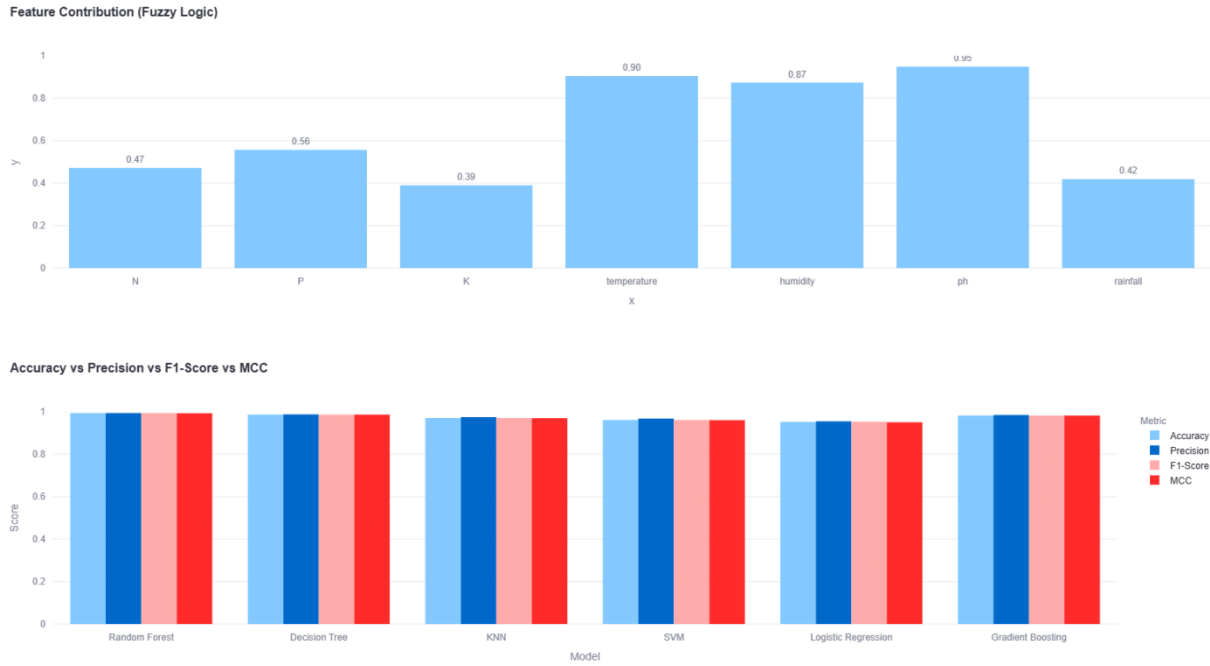


Figure 4. Interactive Plotly data visualizations showing Feature Contribution and the Comprehensive ML Metrics Comparison.

7. Conclusion and Future Scope

The proposed Hybrid Crop Recommendation System successfully bridges the gap between unpredictable agricultural data and clear, transparent digital advising. By implementing Triangular Membership Functions, the system natively handles the continuous nature of soil and climate variables, effectively eliminating the brittleness of standard Boolean logic. The integration of transparent feature breakdowns alongside traditional Machine Learning classifiers proves that algorithmic clarity does not have to come at the expense of computational capability.

Future Scope: Future iterations of this project could integrate real-time IoT soil sensors and live API weather data directly into the web application, entirely automating the input phase and creating a continuous monitoring system for farmers.